Day 3: mathematical diagnostic

Name: _____ Date: _____

60 minutes total. No calculator, notes, internet or AI on the independent pass. Show your reasoning. If stuck, leave the useful work you have and move on. This is not a speed contest.

3 minutes setup · up to 30 minutes core · up to 20 minutes for two selected follow-ups · 7 minutes for your log. The tutor will supply follow-ups. Do not work through the whole reserve book.

If you finish a block early, check your work or continue to the next block with the tutor. Do not exceed the overall limit.

I1-3

Solve $(x-1)/(x+2) \geq 0$ over the reals.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I2-3

Find all real x satisfying $|x-1|+|x-5|=4$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I3-3

Positive a,b have $a+b=10$. What is the greatest possible ab, and when is it attained?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I1-2

Solve $-3x+2<11$ over the reals.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I2-2

Solve $|2x-3|\le 5$ over the reals.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I3-2

For $x>0$, find the minimum of $x+9/x$ and the x attaining it.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I2-1

For $a>0$, express $|x-c|\le a$ as an interval and explain it as distance.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I3-1

State the two-variable arithmetic–geometric mean inequality, its domain and equality condition.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

Session log

Which question was hardest to start, and what did you try?

_____

Was any question familiar, hinted, interrupted, or affected by outside study? Record IDs and details.

_____

Follow-up IDs attempted: _____ / _____